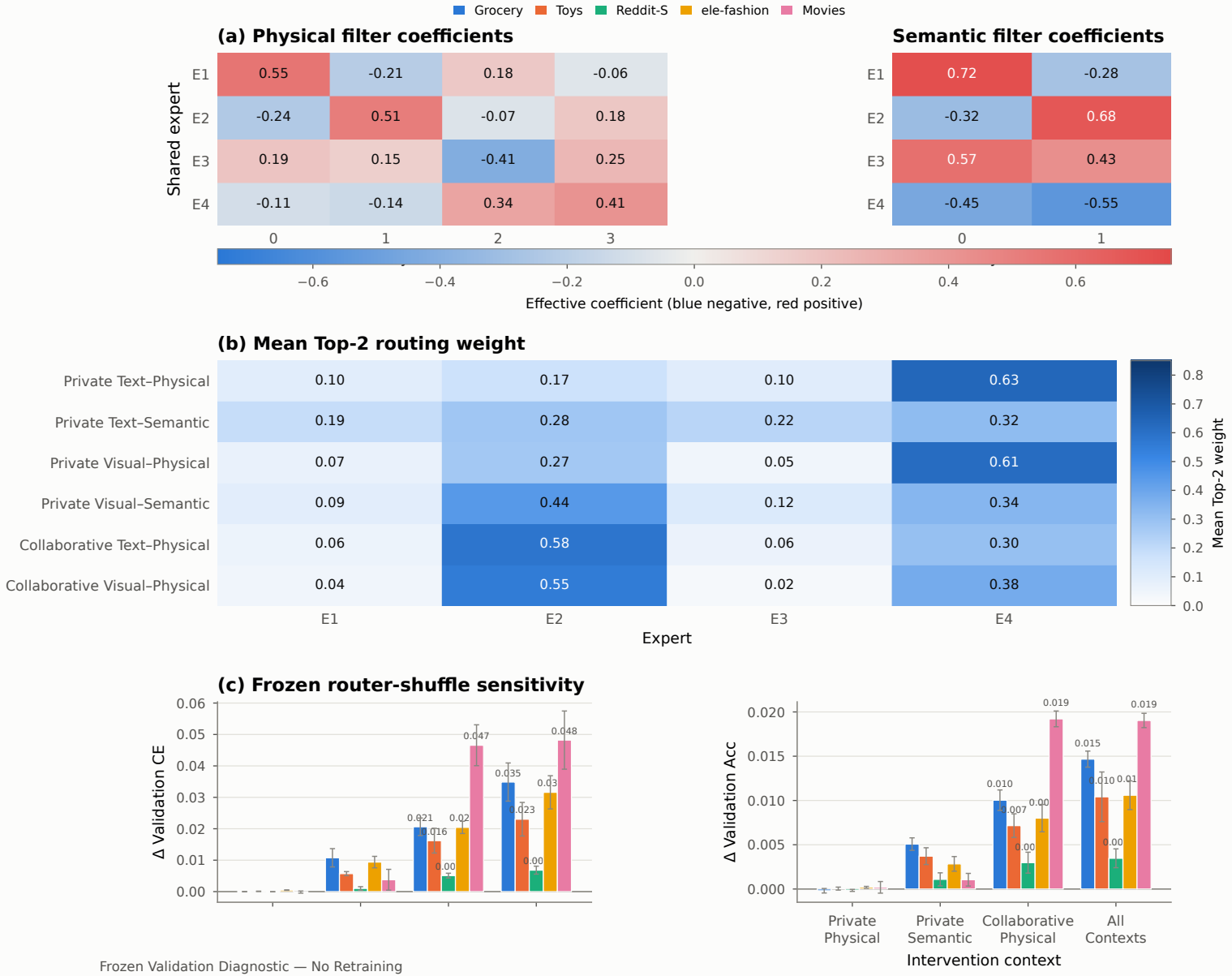
Grocery
Toys
Reddit-S
ele-fashion
Movies
(a) Physical filter coefficients
Shared expert
E1
E2
E3
E4
0.55
-0.21
0.18
-0.06
-0.24
0.51
-0.07
0.18
0.19
0.15
-0.41
0.25
-0.11
-0.14
0.34
0.41
0
1
2
3
Semantic filter coefficients
0.72
-0.28
-0.32
0.68
0.57
0.43
-0.45
-0.55
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Effective coefficient (blue negative, red positive)
(b) Mean Top-2 routing weight
Private Text–Physical
Private Text–Semantic
Private Visual–Physical
Private Visual–Semantic
Collaborative Text–Physical
Collaborative Visual–Physical
0.10
0.17
0.10
0.63
0.19
0.28
0.22
0.32
0.07
0.27
0.05
0.61
0.09
0.44
0.12
0.34
0.06
0.58
0.06
0.30
0.04
0.55
0.02
0.38
E1
E2
E3
E4
Expert
Mean Top-2 weight
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
(c) Frozen router-shuffle sensitivity
Δ Validation CE
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.021
0.016
0.00
0.02
0.047
0.035
0.023
0.00
0.03
0.048
Δ Validation Acc
0.000
0.005
0.010
0.015
0.020
0.010
0.007
0.00
0.00
0.019
0.015
0.010
0.00
0.01
0.019
Private Physical
Private Semantic
Collaborative Physical
All Contexts
Intervention context
Frozen Validation Diagnostic — No Retraining